



SUPERIOR UNIVERSITY

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Assembly Language Project

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Number Guessing Game in 8086 Assembly Language

Git hub link:

<https://github.com/Ejiz123/My-work/tree/main/assembly>

Introduction

This project is a simple Number Guessing Game developed using 8086 Assembly Language. The game generates a random number based on the selected difficulty level, and the user tries to guess the correct number within limited attempts.

Objectives

1. To understand the fundamentals of Assembly Language programming.
2. To learn how DOS interrupts work.
3. To implement loops and conditional branching.
4. To handle keyboard input and screen output.
5. To create a simple interactive game using Assembly Language.

Tools and Technologies Used

- 8086 Assembly Language
- NASM Assembler
- DOSBox
- DOS Interrupts

Project Description

The Number Guessing Game is a console-based game where the computer randomly selects a number according to the chosen difficulty level.

The user selects:

- Easy
- Medium
- Hard

Each level has:

- Different number ranges
- Different attempts

The player enters guesses through the keyboard.

The program compares the entered number with the secret number and displays:

- "Too High!" if the guess is greater
- "Too Low!" if the guess is smaller
- "Correct Guess!" if the guess is correct

After winning or losing, the player can choose to play again.

Difficulty Levels

Difficulty	Number Range	Attempts
Easy	0–9	5

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Difficulty	Number Range	Attempts
Medium	0–50	7
Hard	0–99	10

Features of the Project

- Random number generation
- Multiple difficulty levels
- Input validation
- Too High / Too Low hints
- Play Again feature
- Attempts system
- Single and double-digit support

Working of the Program

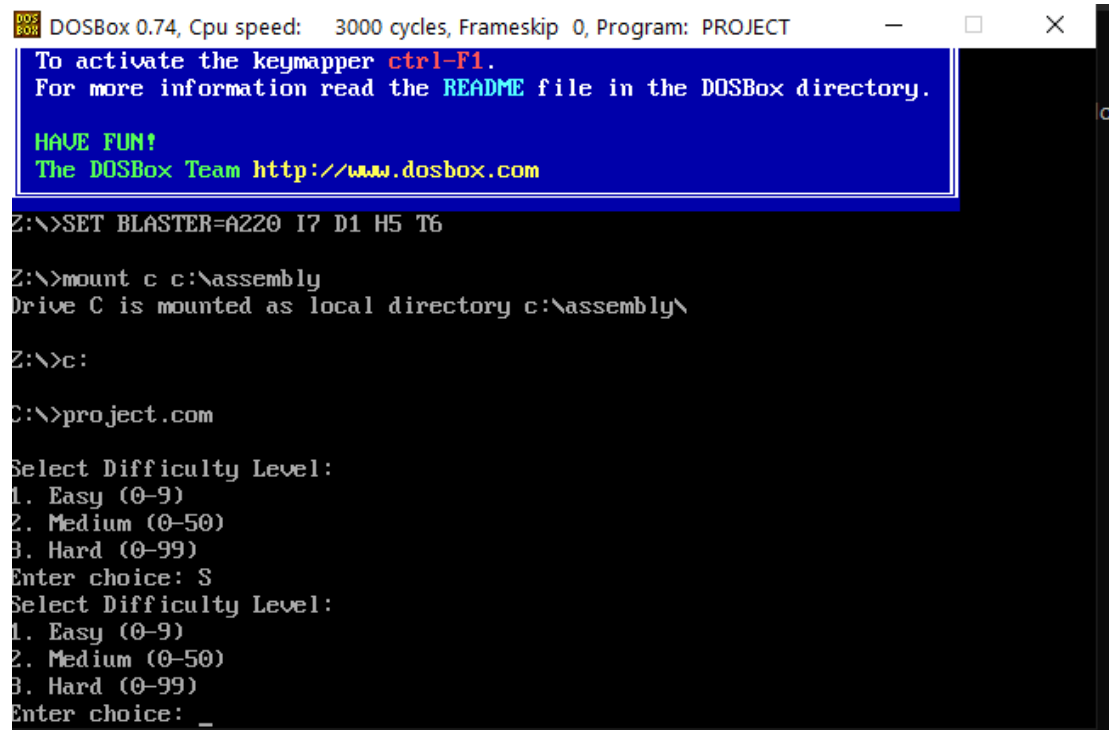
1. Program initializes variables.
2. Ask about difficulty level.
3. Generates a random number.
4. Displays message to user.
5. Takes user input.
6. Validates input.
7. Compares guess with secret number.
8. Displays result.
9. Ends when user wins or attempts finish.

DOS Interrupts Used

21h / 09h → Display string
21h / 01h → Keyboard input
21h / 2Ch → Get system time
21h / 4Ch → Exit program

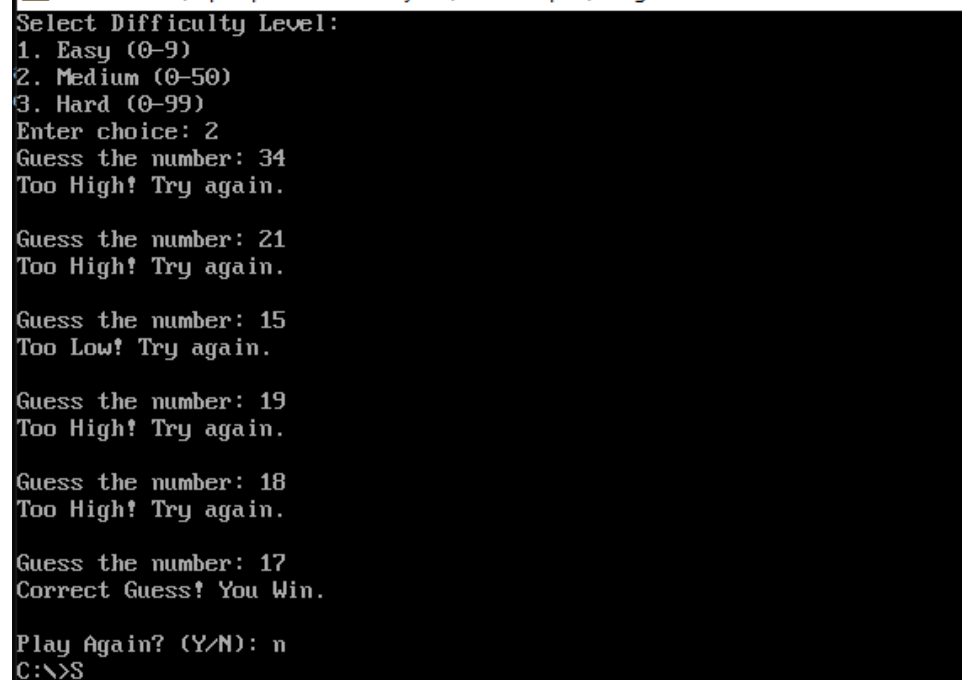
Example run:

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```
DOSBox 0.74, Cpu speed: 3000 cycles, Frameskip 0, Program: PROJECT
To activate the keymapper ctrl-F1.
For more information read the README file in the DOSBox directory.
HAVE FUN!
The DOSBox Team http://www.dosbox.com
Z:\>SET BLASTER=A220 I7 D1 H5 T6
Z:\>mount c c:\assembly
Drive C is mounted as local directory c:\assembly\
Z:\>c:
C:\>project.com
Select Difficulty Level:
1. Easy (0-9)
2. Medium (0-50)
3. Hard (0-99)
Enter choice: S
Select Difficulty Level:
1. Easy (0-9)
2. Medium (0-50)
3. Hard (0-99)
Enter choice: _
```

Output screenshot:



```
Select Difficulty Level:
1. Easy (0-9)
2. Medium (0-50)
3. Hard (0-99)
Enter choice: 2
Guess the number: 34
Too High! Try again.

Guess the number: 21
Too High! Try again.

Guess the number: 15
Too Low! Try again.

Guess the number: 19
Too High! Try again.

Guess the number: 18
Too High! Try again.

Guess the number: 17
Correct Guess! You Win.

Play Again? (Y/N): n
C:\>S
```

Advantages

- Interactive and user-friendly
- Demonstrates Assembly Language concepts

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- Includes multiple difficulty levels
- Supports replay functionality
- Uses random number generation

Future Improvements

The project can be improved by:

- Adding score system
- Adding timer feature
- Displaying remaining attempts
- Adding colorful text
- Creating graphical interface
- Supporting multiplayer mode

Conclusion

The Number Guessing Game with Difficulty Levels successfully demonstrates important 8086 Assembly Language concepts such as loops, conditional jumps, interrupts, input validation, arithmetic operations, and memory handling.

The project provides practical experience in low-level programming and user interaction. The addition of difficulty levels and replay functionality makes the project more advanced, interactive, and suitable for a semester-level Assembly Language project.
